

Typed Decisions in Agent Memory: Where They Help, Where They Don't, and What It Costs

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Abstract

Typed decisions cut an agent memory system's decision cost and latency and improve retrieval under a small budget; they do not change answers after facts change, and a small model used zero-shot cannot make them. Memory systems make an LLM call for every write decision, which makes revisiting the store unaffordable, yet most of that work is choice among fixed options. In engram, an LLM extracts facts and every decision is a typed question answered by a hosted decision model (Jev). With extraction held identical to mem0 2.1.0's, typed decisions cut decision-layer cost by $70.0\times$ and median decision latency by $27.6\times$, with equal dev accuracy (31/35 against 31/35). On 610 held-out LoCoMo questions under a three-memory retrieval budget, engram answers +8.7 points more than mem0 at matched context tokens (95% CI +5.2 to +12.1); at $k=20$ the systems tie (+0.8, CI -2.3 to +3.9). A belief-state store policy over-closes no labeled keep item. Two results are negative: closing stale facts does not change answers on current benchmarks, and the 421M base checkpoint of the open-weights decision model Laya, used zero-shot as its documentation advises against, does not make the relational decisions.

1 Introduction

The cost problem. A memory system for an LLM agent turns conversation into facts it can retrieve later. Each write has two parts:

- Extraction: what facts does this message state?
- Decisions: is each fact new, a duplicate, a refinement, or a change to something already stored? Is it worth keeping? Is it sensitive?

In current open-source systems both parts are LLM calls. mem0 2.1.0's default `add()` makes a single LLM call per message with its additive extraction prompt, and has no separate update or delete step (mem0/memory/main.py, `Memory._add_to_`

`vector_store`). An LLM call per decision is expensive enough that nothing re-examines the store afterwards. Facts that stopped being true stay in it.

The observation. The decisions are choices among options fixed in advance. That is the setting typed decision models are built for. They return a probability over a fixed option set in one short request, and they cost far less than generating text.

Concurrent work. The architectural idea of routing memory decisions to a typed decision model was reached independently by Jev-Mem (Jiang et al., 2026a), which uses Jev for typing, relation construction, query routing, budget allocation, traversal, candidate scoring and stopping. It reports a LoCoMo judge score of 0.777 with a 158 s build and 0.93 s query against A-MEM (Xu et al., 2025), MAGMA (Jiang et al., 2026b) and two further systems reported in that paper. It does not isolate the decision layer, evaluate updates or closes, measure calibration, or use a held-out split or confidence intervals. Separately, a community article reranked AtMem's top-10 with Jev on 1,986 LoCoMo questions and measured the effect at the ranking level: $\text{MRR}@5$ rose from 0.4259 to 0.5868 and $\text{Recall}@1$ from 0.3399 to 0.5423 (Taghia, 2026). This paper's contribution is the controlled measurement: an identical-extraction ablation of the decision layer, its calibration, the safety of the store it drives, the rerank effect on answers with a matched-context control, and the negatives.

Contributions.

1. With extraction held identical, typed decisions replace an LLM decision layer at $70.0\times$ lower cost and $27.6\times$ lower median latency with no measured accuracy loss (§5.1).
2. A belief-state store policy with reversible, gated closes over-closes no labeled keep item, and its v3 rule removes a weak-evidence failure that closes true facts (§3.3, §5.3).

3. Under a three-memory budget a listwise Jev rerank puts engram +8.7 points ahead of mem0 at matched context on 610 held-out questions, and the systems tie at $k=20$ (§5.2).
4. Closing stale facts does not change answers on current benchmarks, and a zero-shot base checkpoint of an open-weights decision model does not make the relational decisions (§5.4).

Scope. We compare against one baseline (mem0 OSS 2.1.0) on one benchmark (LoCoMo: one development conversation and four held-out conversations, scoring four of its five question categories; adversarial is excluded, following mem0’s evaluation protocol). We use update sets written by the system’s author, and one hosted decision model (Jev, jev-1.13.0).

We did not test:

- Zep/Graphiti or Letta
- LongMemEval
- other extraction, answer or judge models
- multi-user stores
- non-English text

2 Background and Related Work

2.1 Write decisions in existing systems

mem0 (Chhikara et al., 2025) extracts and consolidates facts with LLM calls. Version 2.1.0 extracts with one LLM call per message; the prompt receives the new message, the last 10 messages and the 10 most similar existing memories, and emits only additions. Its older update step, which asked an LLM to label each new fact as ADD, UPDATE, DELETE or NONE against existing memories, is still shipped as DEFAULT_UPDATE_MEMORY_PROMPT. We use it as the LLM decision layer in §5.1.

Other systems also decide with an LLM. Zep’s Graphiti builds a temporal knowledge graph and resolves entities and invalidates edges with LLM calls (Rasmussen et al., 2025). MemGPT, now Letta, manages memory tiers through LLM function calls (Packer et al., 2023). A-MEM links notes with LLM-written attributes (Xu et al., 2025), and MAGMA organizes memory as multiple graphs (Jiang et al., 2026b). ByteRover curates a hierarchical context with an LLM; its retrieval is a five-tier progressive strategy that answers most queries in under 100 ms without LLM calls and escalates to agentic reasoning only for novel questions (Nguyen et al., 2026).

2.2 Typed decision models

A typed decision model takes a shared *state* (JSON) and a set of questions. Each question is one of:

- a choice among named options, each with a rubric
- a yes/no (“noul”) question
- a score

It returns a probability distribution per question.

Jev. We use Jev through TypeSafe’s API: model jev-1.13.0, priced at 0.042 USD per million input tokens as billed to our account (src/engram/config.py); the public documentation does not list a price. Its documentation gives a limit of 255 options per choice question (TypeSafe AI, 2026). Measured latency is flat in request size (§5.5).

Laya. Laya is an open-weights model with the same request format. We ran checkpoint convaiinnovations/laya locally through the laya-mlx port on an Apple M2 Pro with 32 GB. It reads at most 512 tokens per question, of which at most 192 go to the instructions and options. This is the 421M base checkpoint. Its model card reports zero-shot typed-decision accuracy of 0.362 against a 0.318 random baseline, calls it “a fast base to specialise, not a zero-shot decision engine,” and offers a separate checkpoint fine-tuned for typed decisions (Convai Innovations, 2026). We used the base checkpoint zero-shot.

2.3 LoCoMo and its limits

LoCoMo (Maharana et al., 2024) contains long multi-session conversations with questions in five categories, of which we score four (adversarial is excluded, following mem0’s evaluation protocol; §4.1). LongMemEval (Wu et al., 2025) also evaluates long-term conversational memory; we used only LoCoMo.

It barely tests updates. In the first 215 messages of conv-26, an LLM labeler (claude-sonnet-4-6, prompt in bench/stale.py) found 2 claims that a later message makes no longer true (bench/slices/conv26_superseded.json).

Public scores are not comparable. Published LoCoMo scores for the same systems differ between the systems’ own reports and third-party reports (Mem0, 2026; ByteRover, 2026). We do not quote those numbers. All comparisons here run both systems under one protocol.

2.4 Closest work and adjacent ideas

Jev-Mem and AtMem. Jev-Mem (Jiang et al., 2026a) is the closest design: Jev controls typing, relation construction, query routing, budget allocation, traversal, candidate scoring and stopping. The two designs differ in the store and in retrieval. Jev-Mem runs with admission filtering off and preserves every observation, so its store is never updated or closed; engram’s store is governed by a close/belief policy with reversible closes and merges (§§3.2–3.3). Jev-Mem routes queries across multiple views; engram uses a single listwise rerank over cosine candidates plus a query-relation pull (§3.1). The AtMem–Jev article (Taghia, 2026) measured the rerank at the ranking level (Recall@10 unchanged; median batch latency 3.32 s) and reported no answer accuracy or intervals.

Reranking and context. Long contexts are used poorly by language models (Liu et al., 2024), and LLMs rerank candidates well (Sun et al., 2023); spending compute on selecting context rather than adding more of it follows from both, and our rerank result is an instance.

Routers. Routers send queries to cheaper models (Ong et al., 2025; Chen et al., 2024), and small classifiers act as guardrails (Inan et al., 2023). The escalation rule of §3.4 is a router of that kind.

Calibration. Temperature scaling (Guo et al., 2017) and conformal prediction (Angelopoulos and Bates, 2021) give principled thresholds for acting on a model’s probability. The belief update of §3.3 depends on the former.

3 engram

3.1 Pipeline

Write path.

1. An LLM extracts facts from a message. In the experiments this uses mem0’s prompt and inputs; §4.2.
2. Each fact goes to Jev as a single request. The request carries the fact questions and one relation_to_candidate question for each of up to 10 candidate facts.
3. A policy layer and a belief state (§§3.2–3.3) turn the answers into store operations.
4. The store is SQLite plus NetworkX. Facts are edges with validity windows, and every decision is logged with its probabilities.

Read path.

1. Embed the question and take the cosine top-k.
2. Jev reranks the candidates with one yes/no relevance question per candidate, plus a query_relation question that can pull in facts by relation type.
3. Add history: the facts each retrieved fact superseded.
4. Render compact lines for the answer model: the date the fact was said, the fact, and the verbatim source quote.

Figure 2 walks through these stages on a question from the retrieval regression test.

3.2 Decision chain and policy layer

The current chain has 12 questions, listed in Appendix A. Choice questions carry a rubric per option. When a question’s wording changes, its version number is bumped and old arms keep the old version.

Decisions act only through explicit rules (src/engram/pipeline/write.py):

- **Thresholds.** A decision acts at $p \geq 0.85$. A superseding relation (update, contradiction, negates) below 0.60 is escalated to an LLM, using mem0’s update prompt. Anything in between is stored as tentative.
- **Temporal gate.** An old edge can close only if the new fact is current or past: $P(\text{current}) + P(\text{past}) \geq 0.85$. Planned and hypothetical statements never close anything.
- **Cardinality.** Each relation type is single- or multi-valued. contradiction closes only a sibling (same subject and relation) on a single-valued relation. update may also close a multi-valued sibling. negates (the new fact says the old one stopped) may close any edge.
- **Two-phrasing agreement.** The first piece of evidence against an edge must be confirmed by a second phrasing of the relation question (relation_to_candidate_recheck). Otherwise it does not count.
- **Plans.** A dedicated yes/no question, plan_fulfilled, asks whether a new fact reports that a stored plan has happened. At $p \geq 0.85$ the plan closes with reason *fulfilled*.
- **Merges are links.** A duplicate becomes a same_as link, collapsed at retrieval and removable.

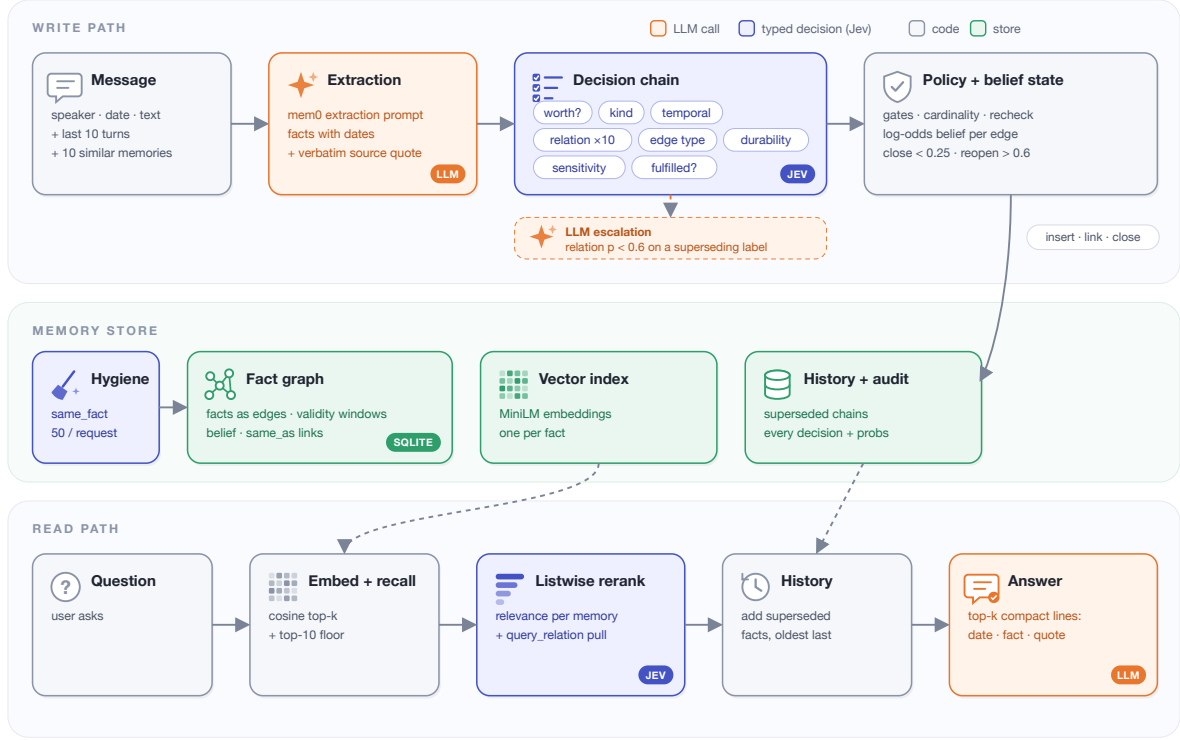


Figure 1: engram’s write and read paths. Orange boxes are LLM calls, blue boxes are typed Jev decisions.

Text is never merged.

- **Overrides.** An explicit request to remember overrides `worth_remembering`. A credential is redacted before storage, whatever the other answers say.

Hygiene. After ingestion, one pass re-checks every group of facts that share a subject and relation. It asks the yes/no question `same_fact` for each pair, 50 pairs per request. At $p \geq 0.85$ the pair is linked, and at $p \leq 0.15$ an existing link is dropped. On the held-out conversations one pass made between 1,865 and 5,446 decisions, for between \$0.0112 and \$0.0332 (§5.3). With LLM decisions, re-examining the store at that scale is what becomes unaffordable.

3.3 Belief state

Each edge i carries a belief $b_i \in [0.02, 0.98]$ that it is currently true. A typed answer z about the edge, with probability q for its chosen label, updates it in log-odds:

$$\text{logit } b_i \leftarrow \text{logit } b_i + w(z) \text{logit } q, \quad (1)$$

$$\text{logit } q \leftarrow \text{logit } q / T. \quad (2)$$

With a per-question temperature T (§5.3), $\text{logit } q$ is first rescaled as in Eq. (2). We use $|w| = 1$. The

sign and the zero cases are exactly the policy rules of §3.2:

- $w = +1$ for duplicate and refinement (support)
- $w = -1$ for update, contradiction and negates where §3.2 permits a close
- $w = 0$ for new, and for any answer a gate blocks

An open edge closes when $b_i < 0.25$ and reopens when $b_i > 0.60$. The gap between the two gives hysteresis.

Belief v3. The v2 rule used in the held-out runs lets any answer count. A duplicate at $q < 0.5$ therefore has $\text{logit } q < 0$ and *lowers* belief, and a weak against-answer raises it. v3 counts an answer only when its label is the argmax and $q > 0.5$ (§5.3). Figure 3 traces one fact from the dev run under both rules.

What this assumes. The log-odds rule Eq. (1) treats q as a calibrated likelihood. Where the model is miscalibrated, belief moves by the wrong amount. This is why §5.3 measures calibration and why a per-question temperature is part of the rule.

3.4 Cost model

With a per-decision Jev cost c_J , an LLM escalation cost c_L and an escalation threshold θ on the top

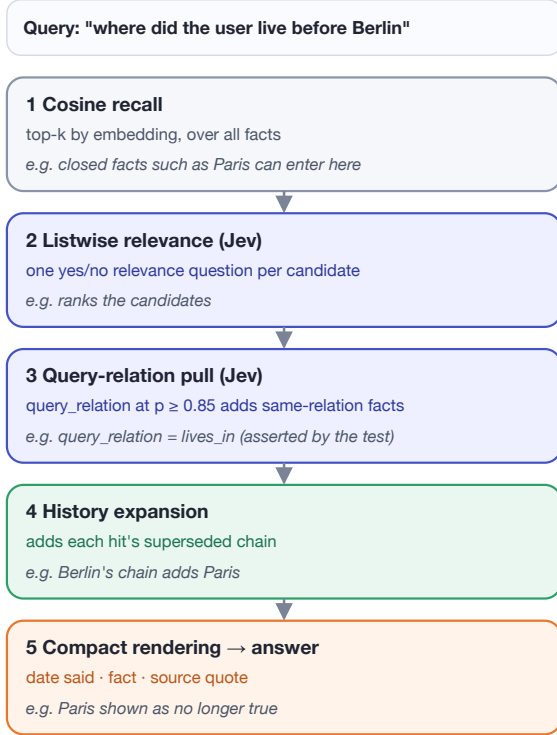


Figure 2: The read path on “where did the user live before Berlin” (fixture in tests/test_retrieval_regression.py). Cosine recall searches closed facts too, so Paris can enter at stage 1 or through Berlin’s superseded chain at stage 4; the test asserts that Paris is retrieved and that Acme, another closed chain, is not, but no stored run records which stage added Paris.

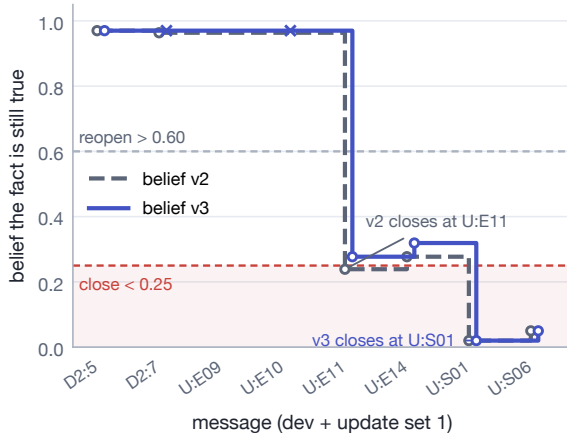


Figure 3: Belief in the fact “Melanie carves out daily me-time through running, reading, or playing violin” (message D2:5), dev + update set 1, under v2 and v3. Under v2 a weak refinement answer at D2:7 ($p = 0.45$) lowers belief slightly, so the update at U:E11 takes it below the close line; v3 ignores that answer (×) and the fact closes one message later, at U:S01. Source: the belief trace in bench/results/e4_belief_v2__dev_updates__k3__noanswer.json and its v3 counterpart.

probability q :

$$C(\theta) = c_J + P(q < \theta) c_L, \quad (3)$$

$$E(\theta) = P(q \geq \theta) \varepsilon_J(\theta) + P(q < \theta) \varepsilon_L, \quad (4)$$

where $\varepsilon_J(\theta)$ is Jev’s error rate on the decisions it keeps and ε_L is the LLM’s error rate on the escalated ones. §5.3 draws the empirical curves of Eqs. (3) and (4).

4 Experimental Setup

4.1 Models, data and protocol

Models (the same in every arm):

- Extraction: claude-haiku-4-5.
- Answers and judging: claude-sonnet-4-6, with mem0’s LoCoMo evaluation prompts (bench/locomo_subset.py).
- The LLM decision layer and all escalations: claude-sonnet-4-6 with mem0’s update prompt.
- mem0 2.1.0 runs on claude-haiku-4-5 with telemetry off.

Slices:

- *dev*: conv-26 sessions 1–4, 76 messages, 35 questions.
- *stress*: conv-26 sessions 1–10, 215 messages, 80 questions.
- *held-out*: conv-30, conv-41, conv-42 and conv-43 whole, 610 questions. These were not used during development. The system configuration was frozen (git tag e4-frozen) before any held-out run.

LoCoMo’s questions fall in five categories, of which we score four (adversarial is excluded, following mem0’s evaluation protocol). Each held-out conversation is ingested once per system, and $k=3$ and $k=20$ are answered from the same store.

Caching and budget. Every LLM and Jev call is cached by its full request, and budget guards stop any run past a spending limit. Phase 2 (all experiments reported here) spent \$93.26 over 80 ledgered runs. Jev accounts for \$1.47 of it (bench/results/phase2_spend.jsonl; per-arm totals in Appendix D). The build phase was not ledgered; roughly \$10 by the author’s estimate.

4.2 Identical extraction

For each message, mem0 2.1.0’s `add()` builds its extraction prompt from:

- the new message, as `[date] speaker: text`
- the last 10 messages
- the 10 existing memories most similar to it
- a current date

It passes no observation date, so relative dates in historical conversations resolve against the current date. We report this as-is, and a patched variant in §6.

engram’s E-arms build the same prompt with the same function (`generate_additive_extraction_prompt`), inputs and model (`bench/run.py`, `src/engram/flags.py`). The “existing memories” differ only because each system’s store differs. The E2 arms differ only in what decides after extraction.

Prompts. mem0’s extraction and update prompts, and the functions that build their requests, are copied verbatim from the installed mem0 2.1.0 package (Apache License 2.0) into `src/engram/llm/prompts_mem0.py`. `tests/test_prompts_mem0.py` checks that the copies are byte-identical to the installed package, so a changed prompt fails the test suite. The answer and judge prompts are mem0’s LoCoMo evaluation prompts, adapted from two per-speaker memory lists to a single list (`bench/locomo_subset.py`).

4.3 Update sets

LoCoMo has almost no updates (§2.3). Two update sets, written in the speakers’ voices, extend it. Both are appended to the dev slice and were frozen by SHA-256 before any run.

Set 1 (`bench/updates_conv26.json`, SHA-256 prefix `2330876388f1`) has 30 items:

- 15 easy closes
- 10 subtle items labeled *no_close*: past-tense mentions and unrealised plans
- 5 fulfilled plans

Set 2 (`bench/updates2_conv26.json`, prefix `d4f3c8d2de61`) adds 28 messages and 20 questions:

- 5 point-in-time questions
- 5 chains of 3–5 changes asked about now
- 5 chains asked about a past moment
- 5 updates with no temporal cue in the message

Appendix B shows one item of each kind; the full sets are in the two files. The system’s author wrote and labeled them, and this is a source of bias.

4.4 Statistics

Both systems answer the same questions, so comparisons are paired: $d_i = \text{engram}_i - \text{mem0}_i$. We report:

- an exact McNemar test on the discordant questions
- a 95% interval from the per-question normal approximation
- a cluster bootstrap that resamples the four held-out conversations. With four clusters, treat it as a robustness check.

The token-matched comparison reports the per-question interval only.

5 Results

5.1 Replacing the decision layer

The three arms in Table 1 share extraction. The two engram arms differ only in what decides after it:

- Jev typed questions (E2 Jev)
- `claude-sonnet-4-6` with mem0’s update prompt, one call per extracted fact (E2 LLM)

The two decision layers answer the same number of questions and mem0 one fewer, within noise. The Jev decision layer costs $70.0\times$ less and has $27.6\times$ lower median decision latency. The LLM decision layer stores fewer facts (18 against 47) because mem0’s UPDATE event rewrites an existing memory instead of adding one.

The two latency columns are medians over different messages. In the E2 LLM arm only 30 of 76 messages made an LLM decision call, so the write median falls on an extraction-only message (median extraction 869 ms); on the messages with decisions, the logged median of the slowest decision call is 7,892 ms (`bench/e2_latency.py`).

Figure 4 shows where each system’s write cost goes: extraction dominates, and the decision layer is 1.3% of the Jev arm’s cost against 47.0% of the LLM arm’s.

These ratios compare Jev against `claude-sonnet-4-6` as the decider. We have no measurement with a smaller LLM decider. On the stress slice the Jev arm scored 67/80 and mem0 64/80; the LLM arm was not run there.

Table 1: Dev slice, conv-26 sessions 1–4, 76 messages, 35 questions, all retrieved memories (default k). Extraction claude-haiku-4-5 for all arms; answers and judge claude-sonnet-4-6; E2 LLM decides with claude-sonnet-4-6. Costs are per 1,000 messages written. Decision p50: median time after extraction, per message, over messages that produced at least one fact. Write p50: median end-to-end write time per message over all messages, including those that produced no facts.

system	accuracy	decision \$/1k	decision p50	total \$/1k	write p50	facts
mem0 2.1.0 (one LLM call per write)	30/35	–	–	\$9.80	879 ms	46
engram E2, LLM decision layer (mem0 update prompt)	31/35	\$8.782	7,675 ms	\$18.70	918 ms	18
engram E2, Jev decision layer	31/35	\$0.125	278 ms	\$9.90	895 ms	47

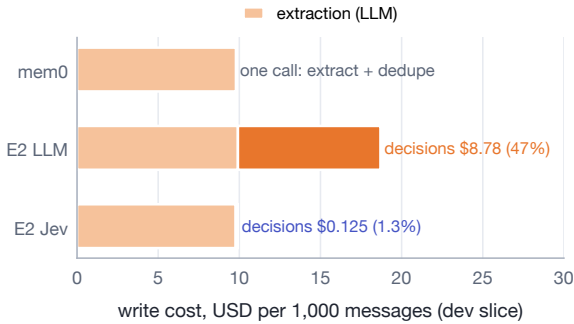


Figure 4: Write cost per thousand messages on the dev slice (76 messages), split into extraction and decision layer. mem0 makes one call per message that extracts and deduplicates.

Held-out parity at default k. The held-out runs compare the frozen full system (E4 belief v2, §3.3) with mem0, not the E2 arms. At $k=20$ engram answers 482/610 and mem0 477/610. The difference is +0.8 points (95% CI -2.3 to +3.9; conversation bootstrap -4.5 to +5.4; McNemar $p = 0.679$), within noise.

5.2 Retrieval under a small budget

Our answer-level result is consistent with the ranking-level improvement AtMem measured when reranking with Jev (Taghia, 2026).

At $k=3$ engram shows the answer model 290 tokens per question and mem0 159. Most of the difference is the source quote on each engram line. To separate context size from ranking, we answered the same 610 questions from mem0’s existing held-out stores at every k from 3 to 8. The token-matched setting is the k whose mean retrieved tokens came closest to engram’s 290: $k=6$, at 315 tokens. That gives mem0 slightly more context than engram. The run cost \$2.52.

At $k=3$ engram is ahead of mem0 by +14.9 points (95% CI +11.2 to +18.7). Against token-matched mem0 the difference is +8.7 points (95%

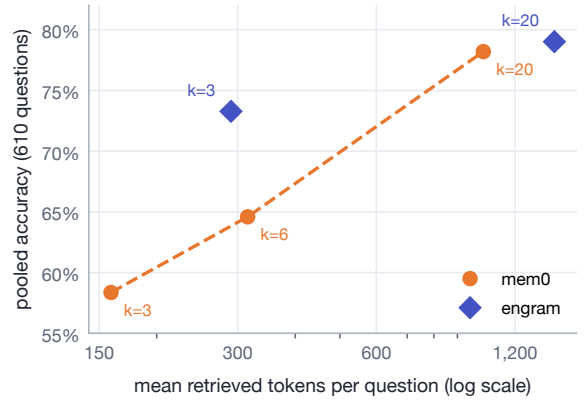


Figure 5: Pooled held-out accuracy (610 questions) against mean retrieved tokens per question. mem0 accuracy was measured at three settings, $k=3, 6$ and 20; the token-matching sweep counted tokens at the other k without answering. Models as in Table 2.

CI +5.2 to +12.1; McNemar $p = 1.3e-06$; 86 questions only engram answered against 33 only mem0 answered).

Figure 5 places the three measured mem0 settings and the two engram settings on one axis of retrieved tokens. engram at $k=3$ sits above the line through mem0’s points, and the two systems meet at $k=20$.

engram’s point estimate is ahead in every conversation and in every category at matched context (Figure 6, Table 4, Figure 7). Per conversation, the matched-context interval excludes zero in 2 of four conversations (conv-42 and conv-43); conv-30 and conv-41 are within noise on their own.

Attribution. Matching context removes 41.8% of the $k=3$ difference. The other 58.2% is what remains once context is matched.

We attribute that remainder to which memories reach the top three, which rests on a dev-slice ablation, not a held-out one. On dev + set 1 at $k=3$ the frozen arm answered 30/30 update questions. It answered 26/30 with Jev reranking off, 30/30 with

Table 2: Held-out LoCoMo accuracy: conv-30, 41, 42 and 43, 610 questions, adversarial category excluded, k as labeled. Extraction claude-haiku-4-5, answers and judge claude-sonnet-4-6. One ingestion per system per conversation. Tokens are mean retrieved-context tokens per question.

system	k	tokens/q	conv-30	conv-41	conv-42	conv-43	pooled
Q			81	152	199	178	610
engram	3	290	72.8%	74.3%	75.4%	70.2%	447/610
mem0	3	159	67.9%	58.6%	55.8%	56.7%	356/610
mem0 (token-matched)	6	315	66.7%	71.7%	59.8%	62.9%	394/610
engram	20	1,461	76.5%	84.2%	78.4%	76.4%	482/610
mem0	20	1,024	77.8%	88.8%	73.4%	74.7%	477/610

Table 3: Paired differences for the rows of Table 2: held-out, Q = 610, k as labeled, models as in Table 2. 95% CI: per-question normal approximation. Bootstrap: resampling the four conversations. Discordant: questions only engram / only mem0 answered correctly.

comparison	Δ (pts)	95% CI	95% CI, bootstrap	discordant	McNemar p
engram $k=3$ – mem0 $k=3$	+14.9	[+11.2, +18.7]	[+8.2, +19.8]	120 / 29	2.4e-14
engram $k=3$ – mem0 $k=6$ (token-matched)	+8.7	[+5.2, +12.1]	not computed	86 / 33	1.3e-06
engram $k=20$ – mem0 $k=20$	+0.8	[-2.3, +3.9]	[-4.5, +5.4]	49 / 44	0.679

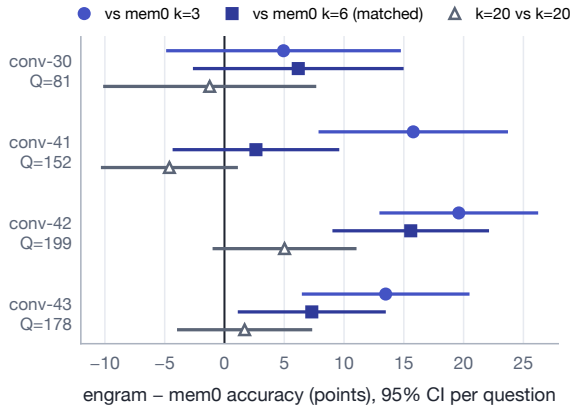


Figure 6: engram minus mem0 accuracy per held-out conversation, with per-question 95% intervals: engram $k=3$ against mem0 $k=3$, against token-matched mem0 $k=6$, and both at $k=20$. Source: bench/results/perconv_diffs.json.

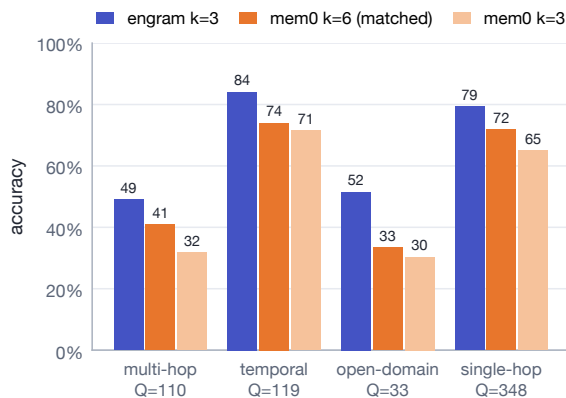


Figure 7: held-out accuracy by category at $k=3$, for engram, token-matched mem0 ($k=6$) and mem0 ($k=3$).



Figure 8: Store outcomes on the update sets: set-1 close items left stale, set-2 stale values, closes on dev + set 1 split by whether they match a labeled pair, and no_close items over-closed. E3 was not run on set 2.

history expansion off, and mem0 answered 25/30. We did not run a held-out no-rerank arm.

5.3 Store safety and calibration

Figure 8 shows the same outcomes per arm.

Over-closes. No labeled keep item was over-closed by any arm.

Closes outside the labels. They separate the policies:

- The first Jev arm (E2), which closed whenever a superseding label cleared 0.85, made 11 closes on dev + set 1. 8 of them match no labeled pair.

Table 4: Held-out accuracy by LoCoMo category, 610 questions, k as labeled. Models as in Table 2.

category	Q	engram $k=3$	mem0 $k=3$	mem0 $k=6$ (matched)	engram $k=20$	mem0 $k=20$
multi-hop	110	49.1%	31.8%	40.9%	67.3%	61.8%
temporal	119	84.0%	71.4%	73.9%	88.2%	84.0%
open-domain	33	51.5%	30.3%	33.3%	51.5%	57.6%
single-hop	348	79.3%	64.9%	71.8%	82.2%	83.3%

Table 5: Store safety on dev + update set 1 and set 2 (storage metrics only). Extraction claude-haiku-4-5, decisions Jev (jev-1.13.0). Columns: set-1 no_close items over-closed / stored; set-2 keep items kept / stored; closes on dev + set 1, split into those matching a labeled pair and those not. Over-closed: a labeled no_close item whose fact was closed by its own update message. Matching a labeled pair: a close whose (closed fact’s message, closing message) is a labeled update or superseded pair. The set-2 keep set has one item.

arm	set-1 over-closed	set-2 kept	closes	labeled	unlabeled
mem0	0/7	1/1	0	0	0
E2: Jev, replace-on-update	0/8	1/1	11	3	8
E3: structural rules	0/8	not run	3	2	1
E4: belief v1	0/8	1/1	3	2	1
E4: belief v2 (frozen)	0/8	1/1	4	3	1
E4: belief v3	0/8	1/1	4	2	2

- Belief v2 made 4, of which 1 match no labeled pair.
- On set 2 the same comparison is 10 of 14 against 1 of 10.

Stale items. A stale item is a close item whose old fact is still active. Belief v2 left 14/18 of the set-1 close items stale, against 17/18 under belief v1 and 18/18 for mem0, which never closes. On set 2 it left 10/16 stale, against 15/16 and 16/16.

Fulfilled plans. A relaxed heuristic closed a plan whenever a related past or current fact arrived. Across its two arms it fired 25 times. 1 firing matched a labeled (plan, fulfilment) pair, and 24 did not. The frozen arm uses the plan_fulfilled question instead; it was asked 234 times on dev + set 1 and closed 2 plans correctly and 1 wrongly.

Belief v3. On the same data, v3 ignored 36 weak answers on dev + set 1 and 39 with set 2 added. It closed the same four facts as v2 on dev + set 1. One of them, D2:5, crossed the close line at a different message (U:S01 rather than U:E11). Since that pair is not labeled, the audit scores v3 at 2 matching and 2 not matching, against v2’s 3 and 1. Stale and over-close counts are unchanged. v3 targets the weak-support failure that closed true facts under Laya (§5.4).

Hygiene. One pass cost \$0.0015 on dev + set 1 (235 decisions). On the held-out conversations it cost \$0.0112, \$0.0254, \$0.0332 and \$0.0323, making 9, 23, 32 and 24 links.

Contradiction regression. The regression set has 50 contradiction pairs, each an old fact, a new fact and a message, labeled with accepted relations and a temporal status (bench/contradiction_pairs.jsonl). Under the current question versions:

- Jev’s relation choice is in the accepted set for 90.0% of pairs (easy 100.0%, medium 93.3%, subtle 73.3%).
- Its temporal status is right for 94.0%.
- The write path’s close rule is right for 76.0%, with 0 false closes.
- Its mean top probability is 0.88 when right and 0.81 when wrong.

Calibration error. We measure expected calibration error (ECE) on two label sets:

- 29 escalation labels, where Sonnet decided a relation Jev was unsure of
- the 50 gold pairs

On the gold pairs Jev’s relation ECE is 0.14 (fitted $T = 1.48$) and its temporal ECE is 0.04. On the escalation labels, which are by construction the cases Jev was unsure of, relation ECE is 0.15. Table 7 has the full comparison, with Laya.

Cost/error tradeoff. The 29 escalation labels are too few for a curve, so Figure 10 uses the 50 gold pairs.

- $c_J = \$0.000016$ is the mean Jev cost of one relation decision over 36,429 held-out decisions.

Table 6: Contradiction regression (50 gold pairs), relation_to_candidate and temporal_status in one request. Laya: convaiinnovations/laya base checkpoint, zero-shot, fp16, on the local MLX server.

backend	relation exact	temporal	close rule	closes	false closes
Jev (jev-1.13.0)	90.0%	94.0%	76.0%	22	0
Laya, Jev wording (truncated)	48.0%	76.0%	32.0%	0	0
Laya, native wording (fits)	38.0%	76.0%	32.0%	0	0

Table 7: Calibration on the escalation labels (dev + update sets, labels by claude-sonnet-4-6) and the gold contradiction pairs; Jev jev-1.13.0, Laya base checkpoint zero-shot. ECE uses 10 equal-width bins on the top choice. T is fitted by NLL per question. The last column is out of sample (2-fold). Relation probabilities fold negatives into contradiction, since the pairs predate negatives.

labels	question	backend	n	top-1 acc.	mean conf.	ECE	fitted T	ECE at T
escalation	relation	jev	29	79.3%	0.92	0.15	2.51	0.10
escalation	relation	laya	29	10.3%	0.47	0.37	5.41	0.12
gold pairs	relation	jev	50	82.0%	0.87	0.14	1.48	0.14
gold pairs	relation	laya	50	28.0%	0.47	0.20	4.22	0.02
gold pairs	temporal status	jev	50	94.0%	0.95	0.04	1.13	0.04
gold pairs	temporal status	laya	50	76.0%	0.72	0.06	0.74	0.07

- $c_L = \$0.0073$ is the mean cost of one escalation over 148 logged escalations.
- No LLM run on the gold pairs is saved, so ε_L is an assumption, plotted at 0 and 0.1.

With Jev alone the error is 10.0%. At $\theta = 0.85$, 32.0% of decisions escalate, the cost is \$0.002368 per decision, and E is 6.0% ($\varepsilon_L = 0$) or 9.2% ($\varepsilon_L = 0.1$). At the production threshold of 0.60, 8.0% escalate and E is 10.0% or 10.8%.

5.4 Negative results

Closes do not change answers.

mem0 never closes anything and leaves every set-1 close item stale (18/18), yet answers 29/30 set-1 questions at default k . The arm that closes most aggressively leaves 10/18 stale and answers 30/30.

The answer model resolves recency itself: extraction writes dates into the memory text, and the compact rendering adds the date each fact was said. Removing all dates, validity and source from the answer context (no-dates column) still leaves every arm we ran that way at 29/30 or higher on set 1. Only the $k=3$ budget separates the systems (25/30 for mem0, 30/30 for engram), and §5.2 attributes that to ranking, not to closes.

Set 2, which asks about chains of changes and past moments, is at its ceiling for every arm. Current LoCoMo-style questions do not reward a correct store.

A small model used zero-shot. Everything in this part concerns the 421M base checkpoint, used zero-shot as its documentation advises against (Convai Innovations, 2026). The checkpoint fine-tuned for typed decisions (reported at 0.766 on its own benchmark) and fine-tuning on our escalation labels were not tested; they are the obvious follow-up.

On the 50 regression pairs (Table 6), Laya chooses an accepted relation for 48.0% with Jev’s question wording and 38.0% with wording rewritten to fit its 192-token question budget. Jev scores 90.0%. Laya never reaches the action threshold, so it closes nothing. Its gold-pair relation accuracy is 28.0% (Table 7). A temperature lowers its ECE but not its accuracy.

With the frozen Jev arm replayed from cache and Laya answering every request on the side, the two gave the same answer on 44.8% of 6,764 decisions and the same action at the 0.85 threshold on 31.2%. On relation_to_candidate they agreed on 5.9% (Table 15, Figure 11).

When Laya decided every question, on dev + set 1 at $k=3$ it scored 24/35 LoCoMo and 26/30 update questions. Jev scored 29/35 and 30/30. It made 23 closes, of which 23 match no labeled pair, and left 43 of 66 facts active. Most of these closes came from weak duplicate and refinement answers below 0.5 lowering belief, the failure belief v3 removes (§3.3).

Hybrid. Laya answered the two high-volume

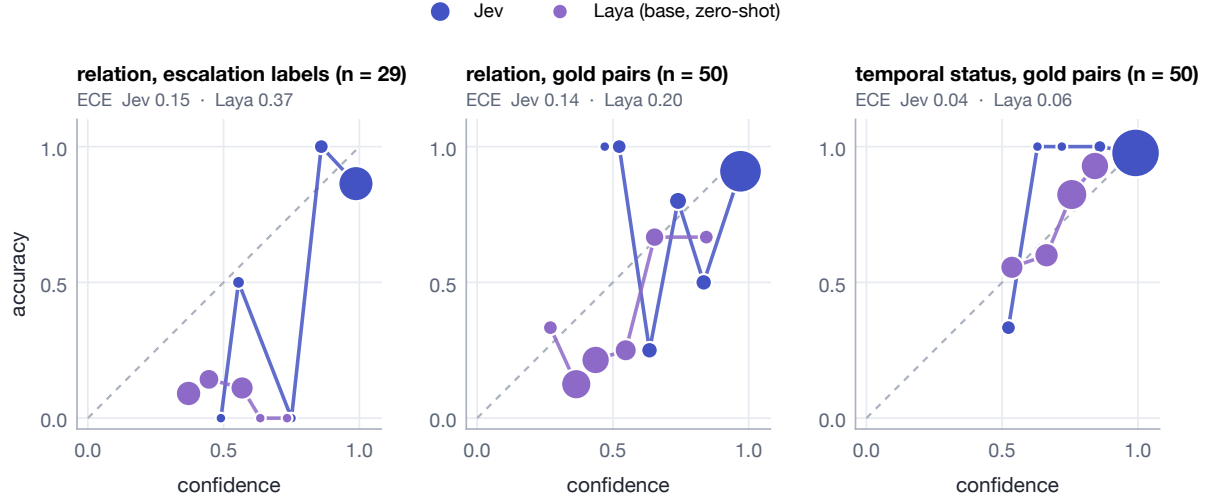


Figure 9: reliability diagrams (accuracy against confidence) for Jev and Laya on both label sets.

Table 8: Update sets on the dev slice (conv-26 sessions 1–4 plus the update messages). Set 1 has 30 update questions and set 2 has 20. Stale counts are close items whose old fact is still active, over close items stored. Accuracy is at default k (all retrieved memories) unless marked. Extraction claude-haiku-4-5, answers and judge claude-sonnet-4-6.

arm	set-1 stale	set-2 stale	set-1 acc.	set-2 acc.	set-1 acc. $k=3$	set-1 acc. no dates
mem0 (add-only)	18/18	16/16	29/30	20/20	25/30	29/30
E2: Jev, replace-on-update	10/18	15/16	30/30	19/20	30/30	30/30
E4: belief v1	17/18	15/16	30/30	19/20	30/30	29/30
E4: belief v2 (frozen)	14/18	10/16	30/30	20/20	30/30	not run



Figure 10: $C(\theta)$ and $E(\theta)$ on the 50 gold pairs. ε_L is assumed, not measured; the 29 escalation labels were too few for a curve.

yes/no questions (relevant_to_query, same_fact) and Jev the rest.

- Agreement on the routed questions: same answer

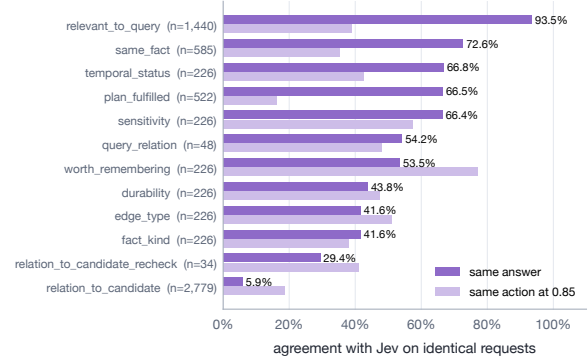


Figure 11: Agreement between the Laya base checkpoint (zero-shot) and Jev on identical requests, per question, sorted by same answer. Dev + update sets 1 and 2, frozen arm's trajectory.

86.9%, same action 34.1%.

- Writes were identical to the all-Jev arm. At $k=3$ only 51.8% of Jev's top-3 lines survived.
- It saved 42.6% of Jev's cost on dev + set 1 (\$0.025 per run).
- Table 9 has accuracy.

Table 9: Hybrid against all-Jev: dev slice with update sets (Q per row: 35 LoCoMo, 30 set 1, 20 set 2), k as labeled, extraction claude-haiku-4-5, answers and judge claude-sonnet-4-6.

questions	all-Jev $k=3$	hybrid $k=3$	all-Jev $k=20$	hybrid $k=20$
update set 1	30/30	27/30	30/30	30/30
LoCoMo dev	29/35	29/35	31/35	31/35
update set 2	19/20	17/20	20/20	20/20

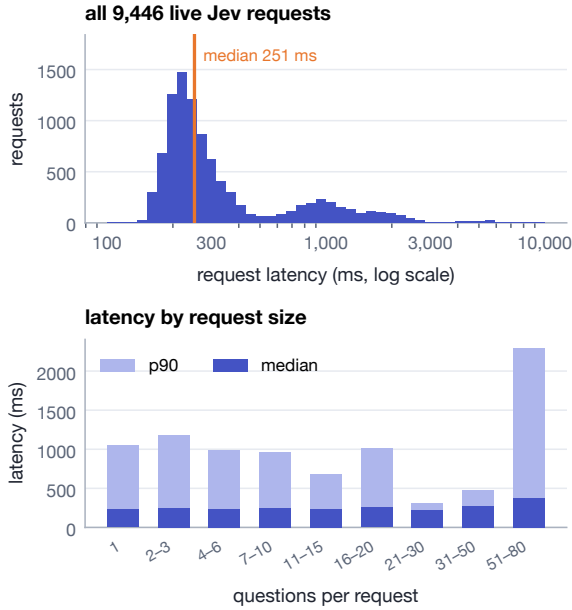


Figure 12: Jev request latency: distribution over all logged requests, and median and p90 by request size.

5.5 Systems notes

Jev latency is flat in request size. Across 9,446 live Jev requests from 25 runs, median latency is between 222 ms and 269 ms for requests of 1 to 50 questions, and 371 ms for 51–80. A least-squares fit gives 380 ms plus 7.19 ms per question (Figure 12; Table 16 in Appendix C). Variation over time is larger than variation over size. For requests of 16–20 questions, the per-run median was between 212 ms and 267 ms in 12 runs, and 793 ms and 607 ms in the two held-out runs made during one slower period.

Extraction dominates write cost. On the held-out set, engram’s write cost is \$10.22 to \$10.55 per 1,000 messages and mem0’s is \$9.92 to \$10.16. The decision layer is 3.3% to 4.8% of engram’s (Table 17, Appendix C).

6 Discussion and Limitations

What the decision layer buys.

- Cost and latency (§5.1).

- An audit trail: every decision has probabilities and a version.
- Store operations that are reversible and gated (§5.3).
- A price at which re-examining the store is routine (§3.2).

What it does not buy.

- Accuracy at default k: it ties (§5.1).
- Update-question accuracy on today’s questions: every arm is near its ceiling (§5.4).

Benchmarks do not see storage correctness. A question about a fact that changed is answerable from a store that kept both versions, as long as the memories carry dates. An evaluation that rewards a correct store would need:

- questions scored against validity windows
- small retrieval budgets, where a stale fact displaces a current one
- memories stored without dates
- long chains of changes

Update set 2 targets these, and every arm is at its ceiling on it.

Rerank or larger k. Under a three-memory budget, a cheap listwise rerank is worth +8.7 points over mem0 at matched tokens. At $k=20$ the ranking stops mattering. For deployments that pay per context token, reranking is the cheaper route to the same accuracy.

Limitations.

- *One baseline and one benchmark.* We compare only mem0 OSS 2.1.0, on four held-out LoCoMo conversations.
- *Author-written update sets.* The update sets and the regression pairs were written and labeled by the system’s author.
- *One decision model.* Jev is the only decision model evaluated as the deciding backend, at one version. Laya was tested only as a zero-shot base checkpoint.
- *mem0’s date handling.* mem0’s open-source path gives no observation date, so relative dates resolve against the run date; the comparison uses it as shipped. A variant with the session date patched in scored 31/35 on dev, against 30/35 unpached, within noise.
- *Dev-only rerank attribution.* The rerank attribution in §5.2 rests on a dev ablation.

- *One judge model*. The judge is an LLM (claude-sonnet-4-6). We did not measure its agreement with human labels.

7 Conclusion

With extraction held identical, typed decisions replaced an LLM decision layer at $70.0\times$ lower cost and $27.6\times$ lower median decision latency, with no measured accuracy loss. On 610 held-out questions, a cheap listwise rerank gave +8.7 points over mem0 at matched context under a three-memory budget, and the systems tied at $k=20$. Closing stale facts, the part of the design aimed at correctness, did not change answers on current benchmarks. The base checkpoint of a small open-weights decision model, used zero-shot, did not make the relational decisions.

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A Decision questions

Table 10 lists the 12 questions in the current chain with their types and options. `edge_type` and `query_relation` choose among 25 relation types. The instructions and the rubric for every option of every version are in `src/engram/decide/questions.py`, and the reasons for each version change are in `docs/DECISIONS.md`.

Table 10: Jev questions in the current chain (`src/engram/decide/questions.py`).

question	type	options
worth_remembering	noul v1	yes / no
fact_kind	choice v1	preference, bio, event, relationship, task, opinion
temporal_status	choice v2	current, planned, past, hypothetical
relation_to_candidate	choice v2	new, duplicate, update, contradiction, refinement, negates
relation_to_candidate_recheck	choice v2	new, duplicate, update, contradiction, refinement, negates
edge_type	choice v1	25 relation types
durability	choice v1	permanent, long_term, short_lived
sensitivity	choice v1	none, health, financial, relationship, credentials
plan_fulfilled	noul v1	yes / no
relevant_to_query	noul v1	yes / no
query_relation	choice v1	25 relation types
same_fact	noul v1	yes / no

B Update sets

Table 11: One item of each kind from the two update sets (bench/updates_conv26.json, bench/updates2_conv26.json).

id	kind	earlier fact or message	update message(s)	question	gold
E01	set 1, easy close	Melanie’s main creative outlet is painting, which relaxes her after a long day.	I haven’t painted in weeks, honestly. Pottery is my main creative outlet now - I go to the studio three evenings a week.	What is Melanie’s main creative outlet now?	Pottery (she has stopped painting)
S01	set 1, subtle (no_close)	Melanie has been married to her husband for 5 years.	Our first apartment after the wedding was so tiny - we still laugh about it.	Is Melanie still married?	Yes, married for 5 years
F01	set 1, fulfilled plan	Caroline plans to continue her education.	I did it - I enrolled in a psychology certificate program at the community college. Classes start next week!	Has Caroline enrolled in a program to continue her education?	Yes, a psychology certificate program at the community college
C01q1	set 2, chain	I just moved into a studio apartment downtown.	Moved again! I’m in a two-bedroom in Oak Park now, so there’s room for a foster kid. → We finally found a house - I live in Evanston now.	Where does Caroline live now?	A house in Evanston
N01	set 2, no temporal cue	My favorite coffee spot is Blue Door Café.	My favorite coffee spot is Grind House.	What is Caroline’s favorite coffee spot?	Grind House

C Per-conversation and systems tables

Table 12: Held-out, per conversation, $k=3$. Q per row; models as in Table 2.

conv.	Q	engram	mem0	Δ (questions)	engram tokens / q	mem0 tokens / q
conv-30	81	59/81	55/81	4	285	154
conv-41	152	113/152	89/152	24	297	161
conv-42	199	150/199	111/199	39	292	163
conv-43	178	125/178	101/178	24	283	153

Table 13: Held-out, per conversation: engram $k=3$ against token-matched mem0 ($k=6$). Q per row; models as in Table 2.

conv.	Q	engram $k=3$	mem0 $k=6$	Δ (questions)
conv-30	81	59/81	54/81	5
conv-41	152	113/152	109/152	4
conv-42	199	150/199	119/199	31
conv-43	178	125/178	112/178	13

Table 14: Held-out, per conversation, $k=20$. Q per row; models as in Table 2.

conv.	Q	engram	mem0	Δ (questions)	engram tokens / q	mem0 tokens / q
conv-30	81	62/81	63/81	-1	1,434	996
conv-41	152	128/152	135/152	-7	1,521	1,049
conv-42	199	156/199	146/199	10	1,491	1,043
conv-43	178	136/178	133/178	3	1,389	995

Table 15: Laya against Jev on identical requests: dev + update sets 1 and 2, frozen arm’s trajectory at $k=3$ (write and retrieval decisions; n per row). Jev jev-1.13.0; Laya base checkpoint, zero-shot. Same action: both choose the same label at $p \geq 0.85$, or neither reaches it. Acts: share of decisions at $p \geq 0.85$.

question	n	same answer	same action	Jev acts	Laya acts
relation_to_candidate	2,779	5.9%	18.5%	79.9%	6.4%
relevant_to_query	1,440	93.5%	38.9%	89.7%	31.3%
same_fact	585	72.6%	35.2%	96.2%	34.4%
plan_fulfilled	522	66.5%	16.1%	90.0%	32.6%
worth_remembering	226	53.5%	77.0%	18.1%	4.9%
fact_kind	226	41.6%	38.1%	63.7%	15.5%
temporal_status	226	66.8%	42.5%	62.8%	5.3%
edge_type	226	41.6%	50.9%	29.2%	55.3%
durability	226	43.8%	47.3%	52.7%	0.0%
sensitivity	226	66.4%	57.5%	45.1%	7.1%
query_relation	48	54.2%	47.9%	39.6%	75.0%
relation_to_candidate_recheck	34	29.4%	41.2%	35.3%	26.5%

Table 16: Jev (jev-1.13.0) latency by request size over all logged runs (dev, stress and held-out slices), from the decision logs (bench/results/jev_latency.json). Client-measured, after the rate limiter, retries included.

questions / request	requests	median input tokens	median latency	p90 latency
1	1,305	564	229 ms	1,052 ms
2–3	1,054	870	248 ms	1,176 ms
4–6	775	1,562	231 ms	992 ms
7–10	1,417	4,314	246 ms	967 ms
11–15	227	2,432	231 ms	680 ms
16–20	3,240	5,593	255 ms	1,009 ms
21–30	184	3,551	222 ms	312 ms
31–50	505	6,903	269 ms	472 ms
51–80	739	9,187	371 ms	2,290 ms

Table 17: Held-out write side (conv-30, 41, 42, 43; writes only). Extraction claude-haiku-4-5 for both systems; decisions Jev jev-1.13.0 with claude-sonnet-4-6 escalations. One ingestion per system per conversation. Decision layer = Jev plus escalations.

conv.	engram \$/1k	decision \$/1k	mem0 \$/1k	decision p50	engram write p50	mem0 write p50	engram facts	mem0 memories
conv-30	\$10.22	\$0.33	\$9.92	1,982 ms	928 ms	1,034 ms	267	276
conv-41	\$10.54	\$0.47	\$10.16	1,614 ms	1,775 ms	1,437 ms	668	822
conv-42	\$10.55	\$0.50	\$10.04	981 ms	1,824 ms	1,300 ms	708	695
conv-43	\$10.45	\$0.50	\$9.96	868 ms	1,819 ms	1,210 ms	628	636

D Reproduction

Each table and figure was produced by the command below, run from the root of the engram codebase. With the call cache (bench/.cache/calls.sqlite) in place, every command replays at no API cost.

The spend ledger records cost per run, not per table. Per-arm totals:

table / figure	command
Table 1; Fig. 4	python -m bench.run --arm {e2_jev,e2_llm,mem0} --slice dev
Tables 2–4, D; Figs. 5–7	python -m bench.run --arm {e4_belief_v2,mem0} --slice heldout:<conv> --top-k 3 --also-top-k 20, then python -m bench.heldout_report and python -m bench.token_match
Tables 5, 8; Figs. 3, 8	python -m bench.run --arm <arm> --slice dev_updates[2] [--top-k 3] [--no-dates]
Tables 7, 6; Figs. 9, 11	python bench/test_contradictions.py --backend laya [--native] --save . . . , python -m bench.calibration, python -m bench.jev_regression (the Laya rows need bench/laya_server.py running)
Fig. 10	python -m bench.tradeoff
Tables 15–9	python -m bench.laya_report, python -m bench.hybrid_report
Table 16, Fig. 12	python -m bench.jev_latency
this paper	python paper/build.py, python paper/figures.py

arm	runs	Jev	Claude
mem0	15	\$0.0000	\$35.94
e4_belief_v2	18	\$1.0917	\$31.80
e2_jev	2	\$0.0636	\$3.51
e2_llm	2	\$0.0134	\$3.21
mem0_token_ matched	1	\$0.0000	\$2.52
e4_belief_v2_ laya	2	\$0.0405	\$2.40
e2_jev_v2	4	\$0.0362	\$2.35
e1_recall	4	\$0.0653	\$2.05
e4_belief	6	\$0.0316	\$1.69
e3_structural_v2	1	\$0.0278	\$1.44
e0_baseline	8	\$0.0736	\$1.36
mem0_dated	1	\$0.0000	\$0.96
e3_structural_v3	2	\$0.0272	\$0.76
e4_belief_v2_ hybrid	4	\$0.0000	\$0.73
e4_belief_v2_ compact	2	\$0.0000	\$0.59
e4_belief_v2_ norerank	1	\$0.0000	\$0.25
e4_belief_v2_ shadow	2	\$0.0000	\$0.18
e3_structural	1	\$0.0008	\$0.04
e4_belief_v2_ nohist	1	\$0.0000	\$0.01
e4_belief_v3	3	\$0.0004	\$0.00